

DEVELOPMENT OF BIOMATERIAL LOADED WITH ANTIMICROBIAL PEPTIDES FOR THE TREATMENT OF ORTHOPAEDIC INFECTIONS

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ABSTRACT

Periprosthetic joint infections (PJIs) are among the most common implant-associated infections, affecting approximately 1-2% of joint replacement patients [1]. These infections can compromise implant success and delay patient recovery. Among the pathogens responsible for these infections, *Staphylococcus aureus* (*S. aureus*) is one of the most common causes of PJIs [2]. It remains one of the most challenging microorganisms due to its ability to adhere to implant surfaces, form biofilms and persist despite antibiotic therapy. Biofilm formation provides a protective barrier that helps bacteria resist to antimicrobial agents [3]. With the increasing threat of antimicrobial resistance, there is an urgent need for antibiotic alternatives. Some antimicrobial peptides (AMPs) are promising candidates due to their antibacterial activity and anti-biofilm activities. Here, GH12, a promising cationic antimicrobial peptide, is used to evaluate its antibacterial and anti-biofilm properties against *S. aureus* [4]. Furthermore, the integration of promising AMPs into biocompatible biomaterials, such as fibrin-based hydrogel, a natural polymer, may provide a strategy for localized and sustained antimicrobial activity while supporting tissue repair [5].

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